Recommendation Scores

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 21: weighted feature contributions are added.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Compute a weighted score. Try the recall before opening notes.

Your learning target

I can compute a **weight**ed **score**.

I can apply hard **constraint**s before **rank**ing.

I can explain how **weight** choices change outcomes.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 23: Recommendation **Score**s

A **weight**ed **score** expresses priorities but also embeds choices.



THINKER MISSION

Create two user profiles and calculate book **rank**ings.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Which is closer under absolute-gap distance: a gap of 2 or a gap of 5?

This week's end goal

By the end, I can **rank** only eligible options with a transparent **weight**ed **score**.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Recommendation Scores case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use recommendation **score**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A fictional activity chooser uses enjoyment and learning ratings from 0 to 5. Its **score** is 2 times enjoyment plus learning. The activity must also take at most 20 minutes.

Duration is a hard requirement; a high **score** cannot compensate for exceeding it.

STUDENT LESSON · 3 OF 18

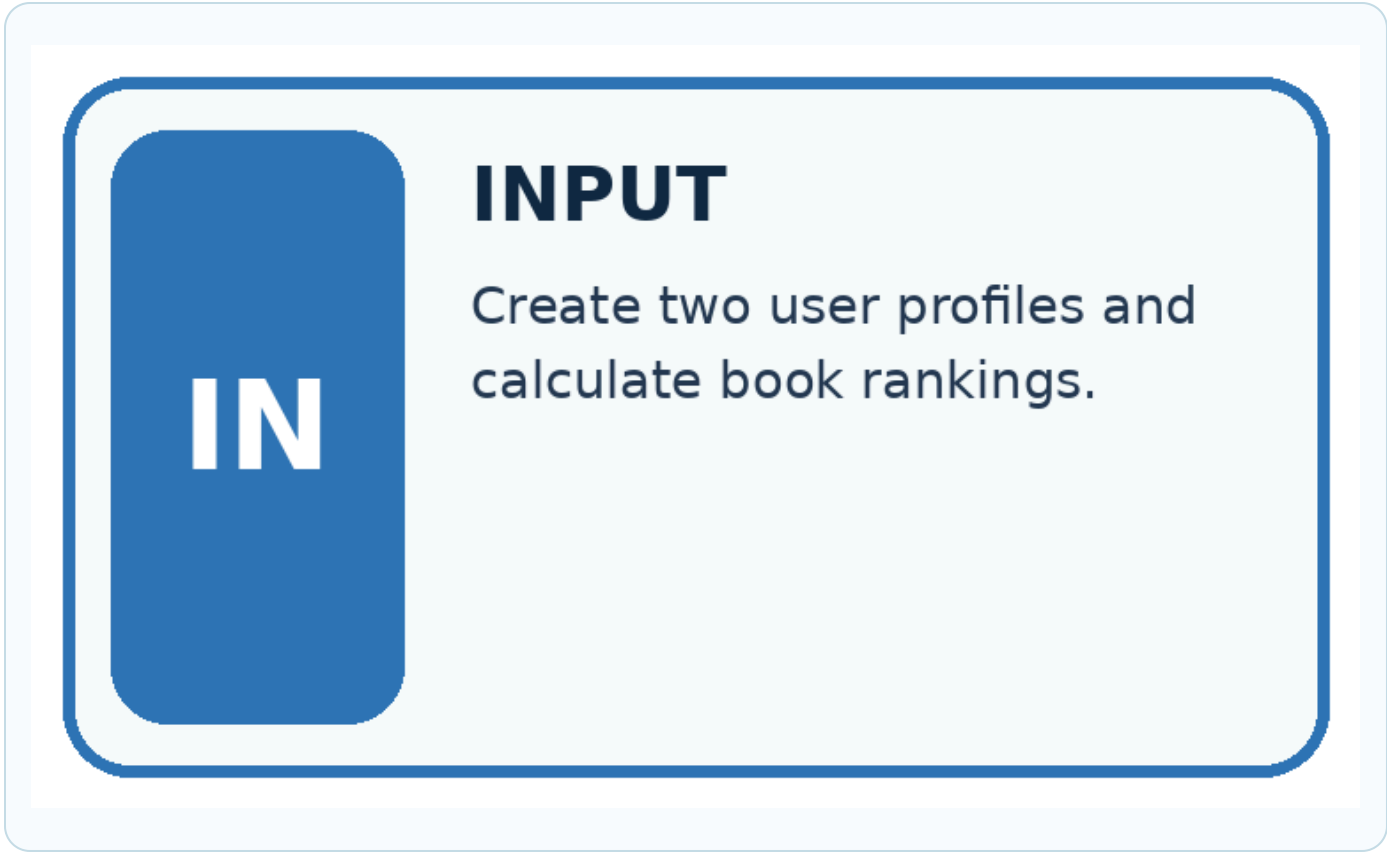
See how it works

> Follow the mechanism for Recommendation Scores. Connect each labelled input, deciding step and result.

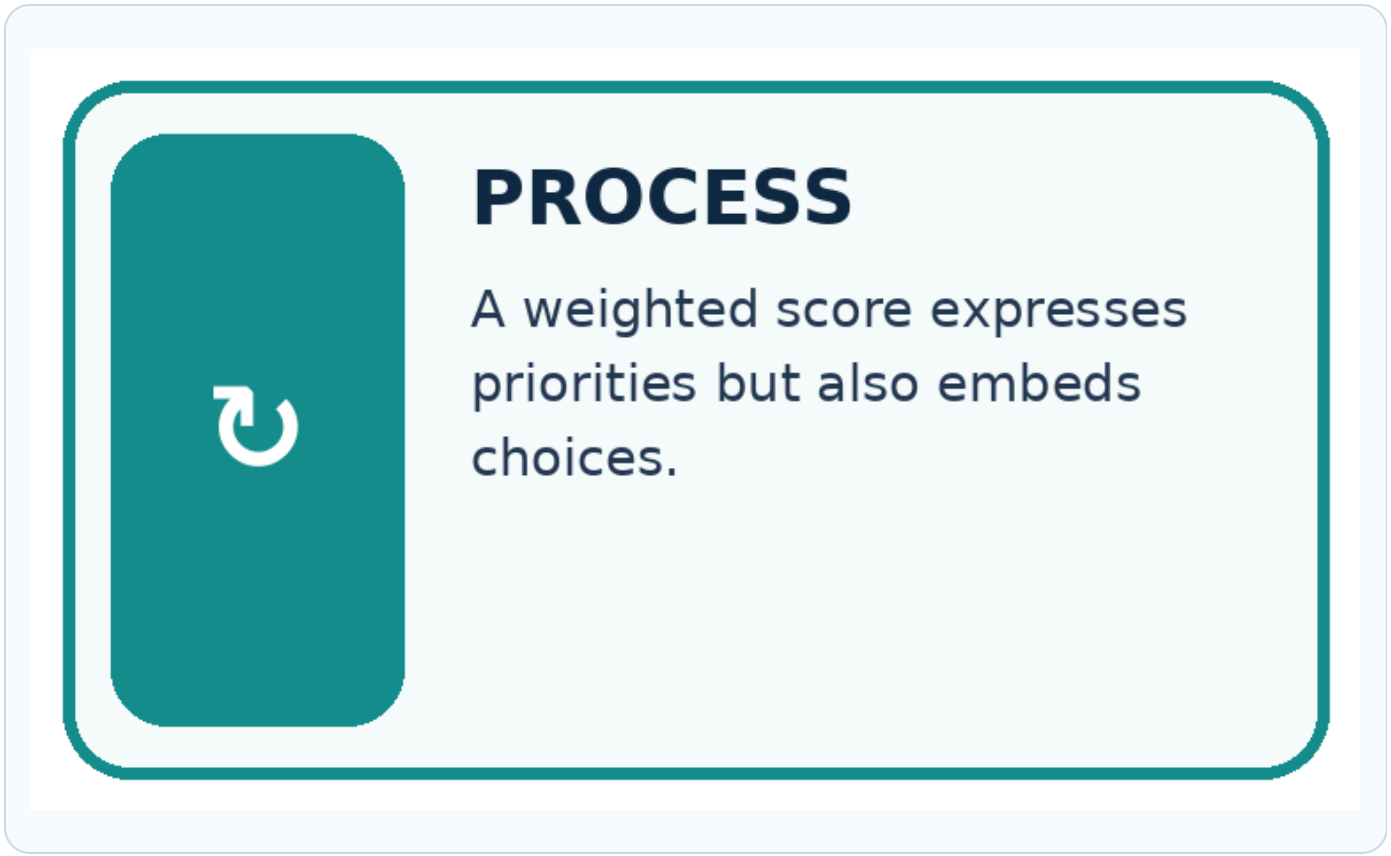
VISUAL MECHANISM

Input → **Process** → **Output**

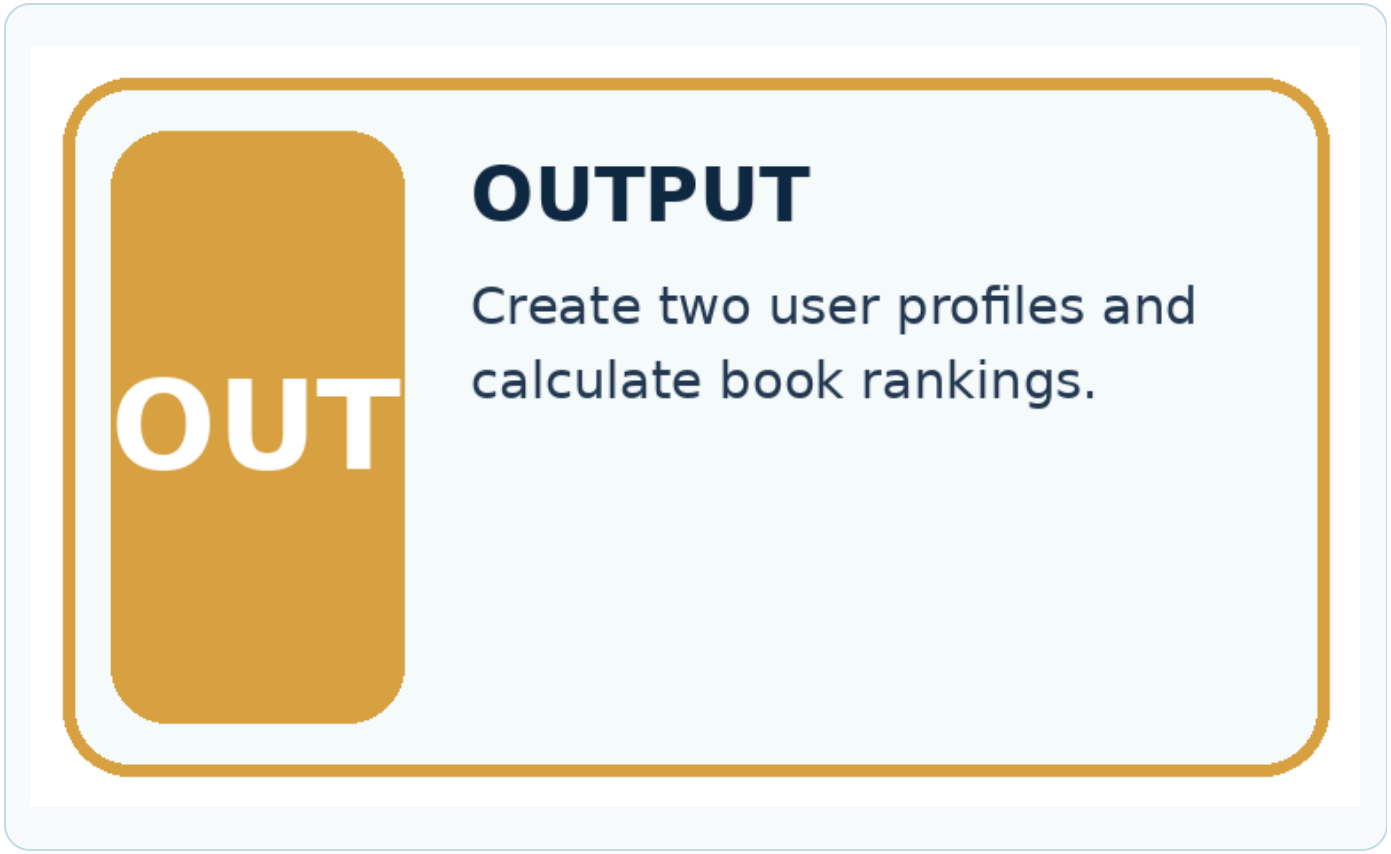
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Create two user profiles and calculate book rankings.

↓


PROCESS
A weighted score expresses priorities but also embeds choices.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

```
options = [{"name": "A", "fun": 4, "learning": 2, "minutes": 15},
           {"name": "B", "fun": 3, "learning": 5, "minutes": 25}]
for item in options:
    if item["minutes"] <= 20:
        score = 2 * item["fun"] + item["learning"]
        print(item["name"], score)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
A	2 * 4 + 2 = 10	Eligible at 15 minutes
B	2 * 3 + 5 = 11	Excluded at 25 minutes

Read each row from left to right and connect the evidence to the stated rule.

Only A is printed, with **score** 10. If B took 20 minutes it would qualify and have the higher **score** 11. The inclusive boundary matters.

Ratings and **weight**s are invented for practice and require justification before use in a real decision.

Check must-have rules before scoring

Make two decisions in order. First check the duration: A takes 15 minutes and qualifies under the 20-minute limit; B takes 25 minutes and does not. Then calculate **score**s for eligible choices. A **score**s 2 times 4 plus 2, or 10.

B would **score** 11, but that **score** cannot remove the time requirement. Keep the rejection reason beside B instead of calling it a worse-scoring activity.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Recommendation Scores. Identify the detail that changes the

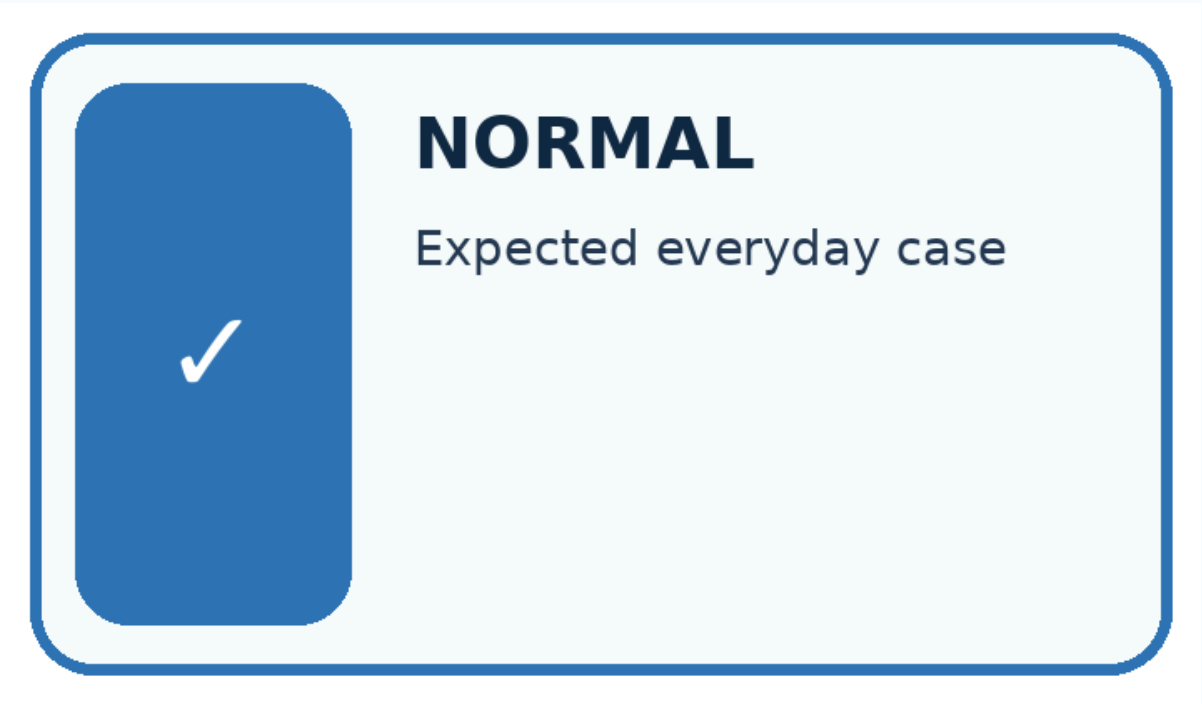
> conclusion.

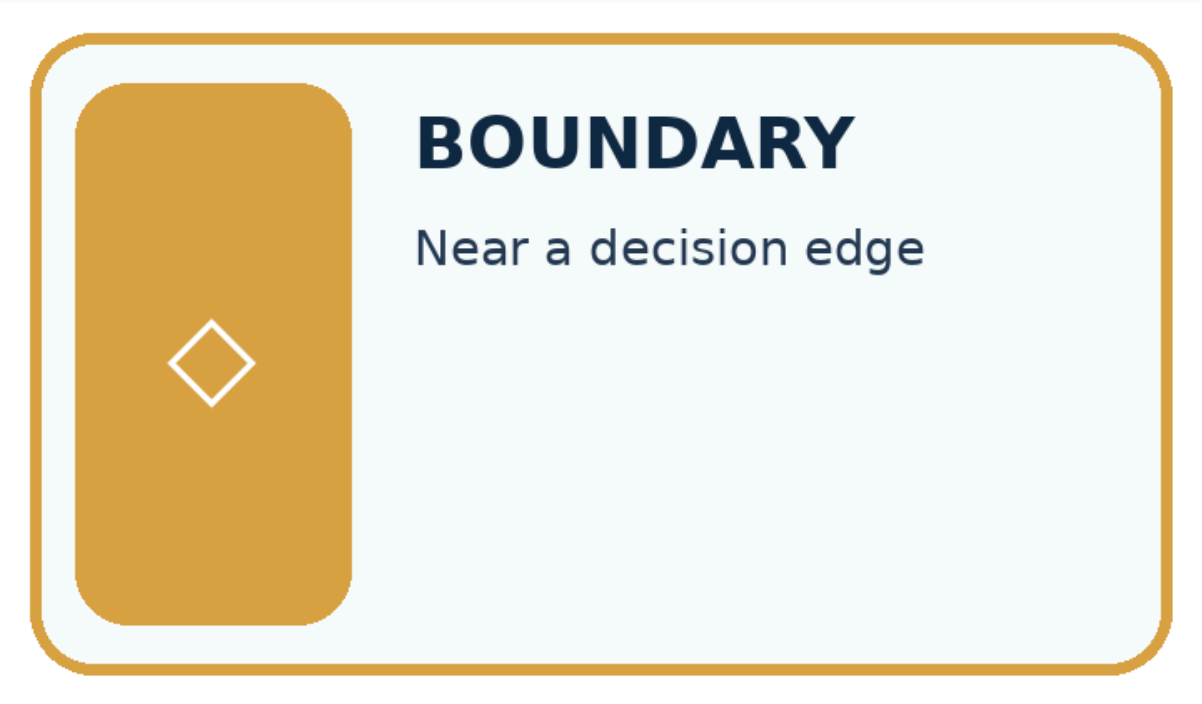
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

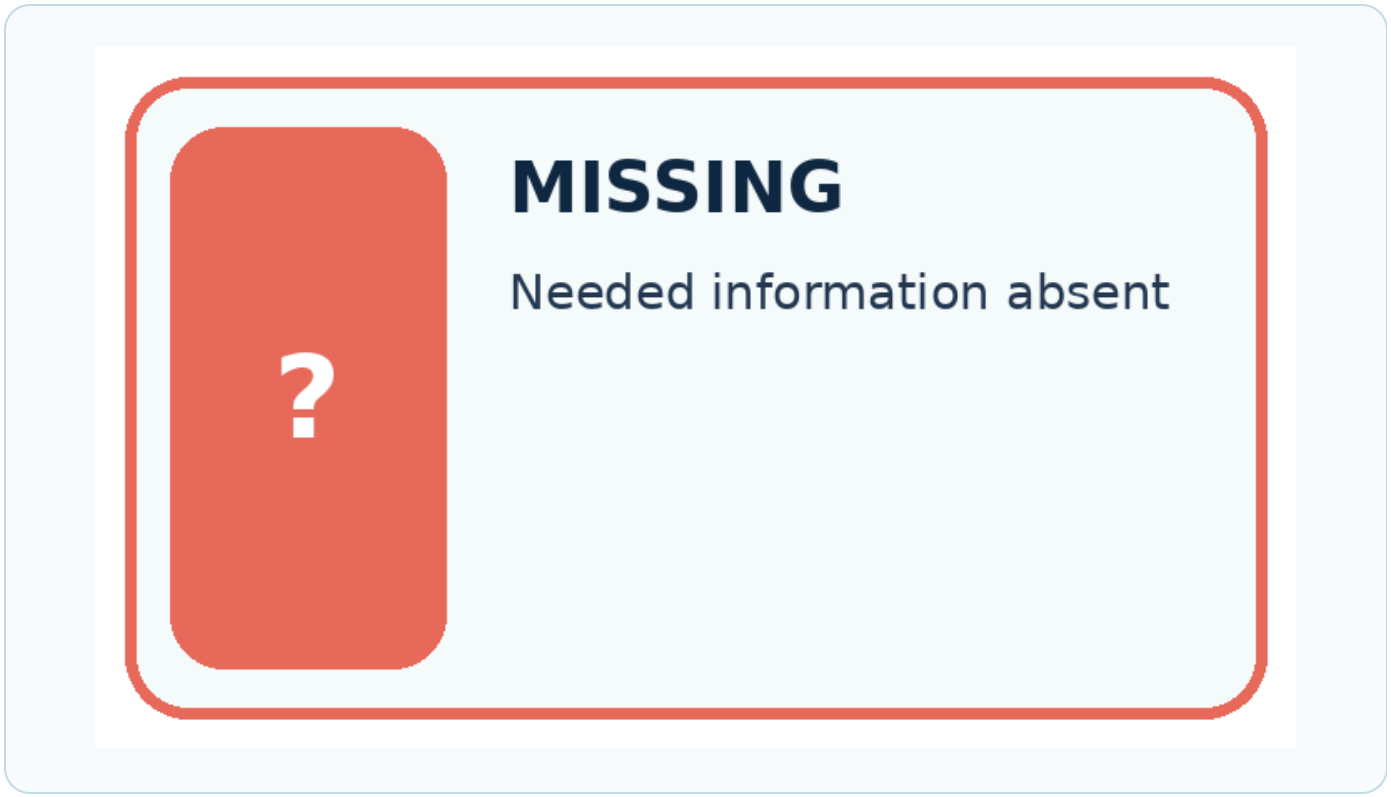
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Change interest **weight** from 2 to 3, keeping time limit 14.

Predicted result

Prediction: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Why the result follows

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

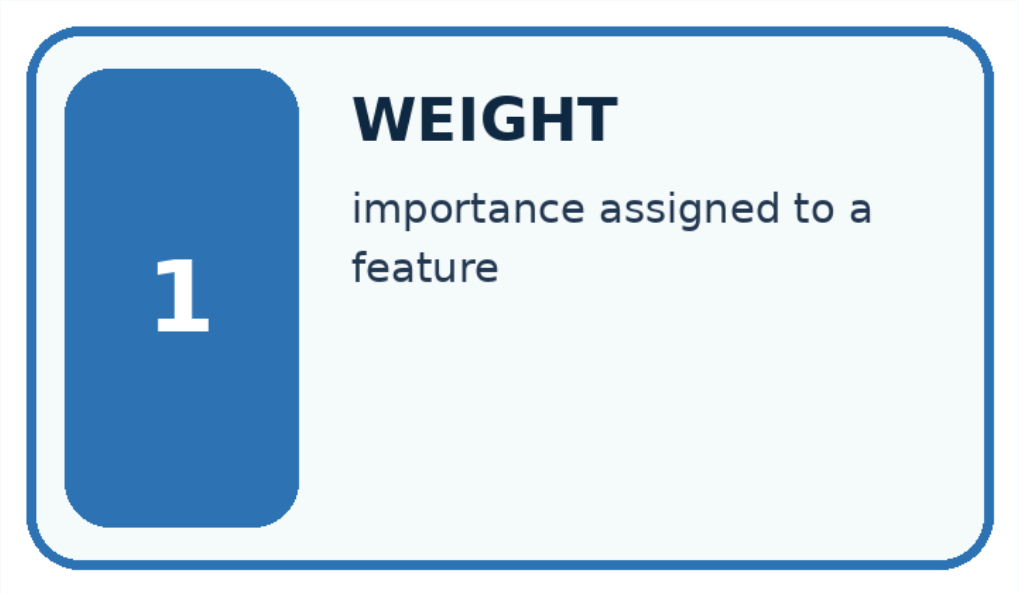
Words that unlock the idea

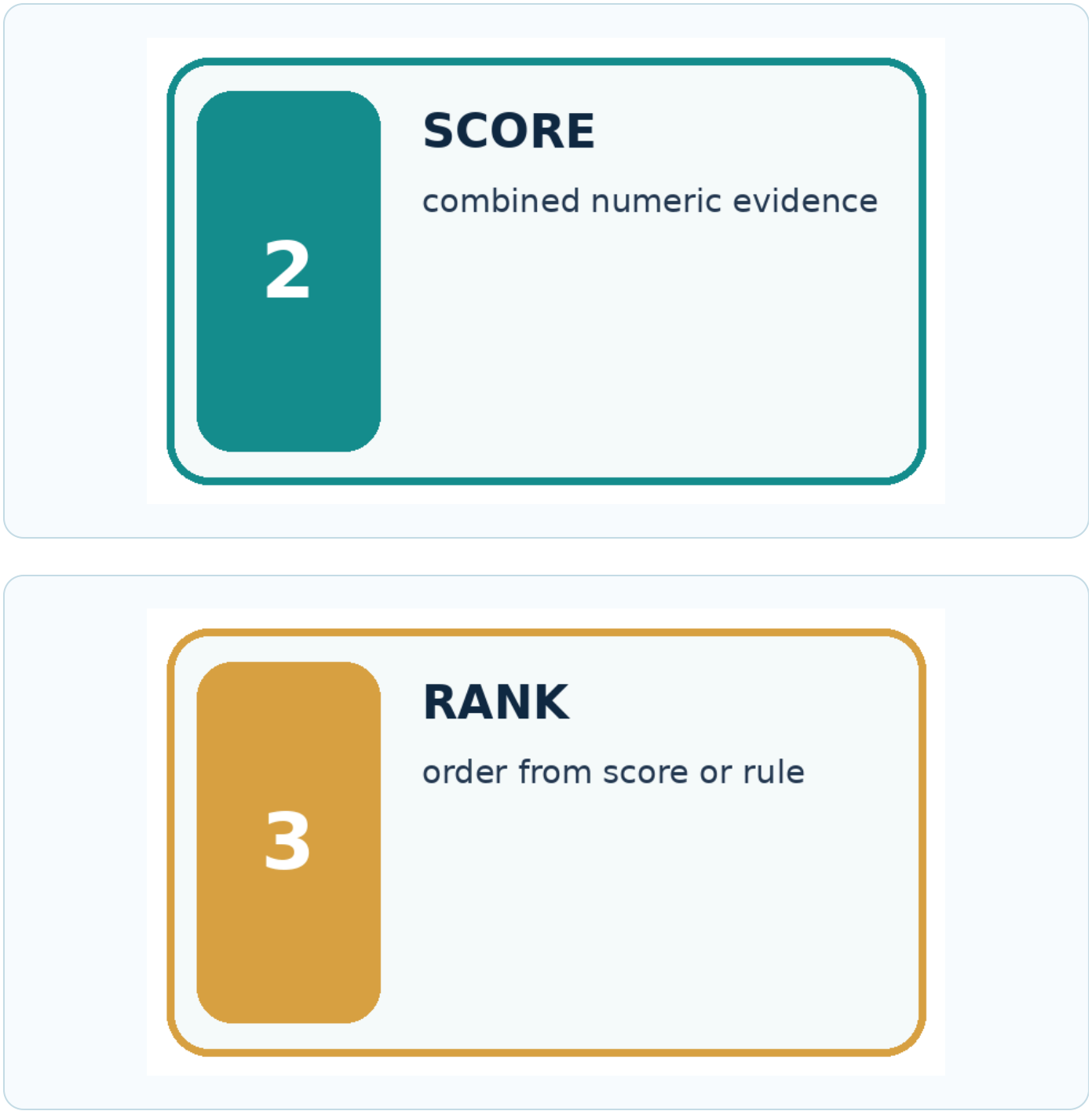
> Use the vocabulary of Recommendation Scores to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
SCORE
combined numeric evidence
3
RANK
order from score or rule



Word	My own example	Not this / boundary
weight	A multiplier controlling how strongly a feature or value contributes to a calculation.	
score	A numerical value assigned using a specified rule; it is not automatically a probability.	
rank	Place candidates in an order using a declared comparison rule.	
tradeoff	A situation where improving one desired property can make another harder to achieve.	

Words used in the closer explanation

Word	Meaning
Constraint	A requirement an option must meet

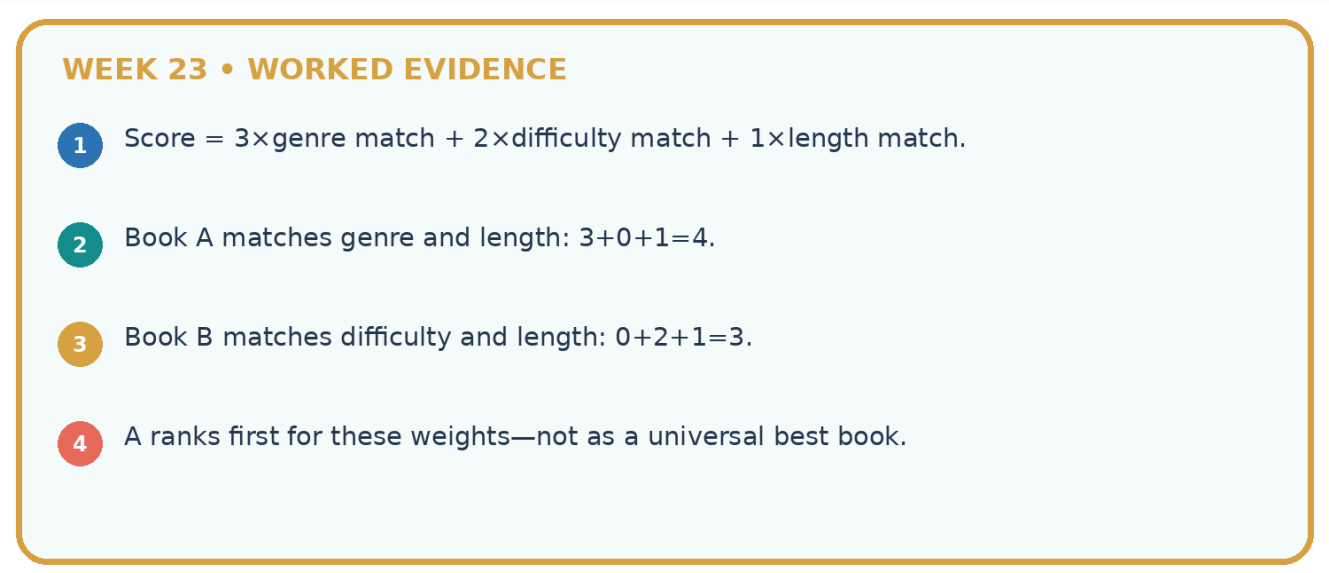
Follow a worked example

> Follow the worked trace for Recommendation Scores, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Score** = 3×genre match + 2×difficulty match + 1×length match.

2. Book A matches genre and length: 3+0+1=4.

3. Book B matches difficulty and length: 0+2+1=3.

4. A **rank**s first for these **weight**s—not as a universal best book.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

If enjoyment weight becomes 1 and learning weight becomes 2, find both scores before applying the same duration limit.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Recommendation Scores

A fictional book-activity chooser has a 14-minute limit.

The facts we will use

Supplied facts

Score=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Supplied facts

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

AI ACADEMY / AI-2/23

Time filtering and scores

	Minutes	Score	Status
A	14	8	eligible
B	15	11	excluded
C	12	9	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/23

Recommendation Scores

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Eligibility first

A and C fit time; B takes 15 and is excluded.



2. Scores

A:2 x 3+2=8. C:2 x 2+5=9.



3. Choose

C wins at score 9 among eligible activities.



4. Explain exclusion

B's score 11 cannot override its failed time requirement.

CHANGE ONE THING

Change interest weight from 2 to 3, keeping time limit 14.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Eligibility first	A and C fit time; B takes 15 and is excluded.
2. Scores	A:2×3+2=8. C:2×2+5=9.
3. Choose	C wins at score 9 among eligible activities.
4. Explain exclusion	B's score 11 cannot override its failed time requirement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

What this example does not establish

Keep the stated limit

Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

Keep the stated limit

Read the labelled picture: Why does B not become the winner when its **score** is largest?

Keep the stated limit

Change one thing: Change interest **weight** from 2 to 3, keeping time limit 14. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Recommendation Scores claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Weights are neutral facts rather than design choices. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Weights represent preferences, not a discovered truth. Explain who chose them and test whether small changes alter the recommendation. Avoid claiming that a larger **score** proves greater educational value.

> **For eligible A (fun 4, learning 1) and B (fun 2, learning 4), compare weights (2,1) with (1,2).**

STUDENT LESSON · 8 OF 18

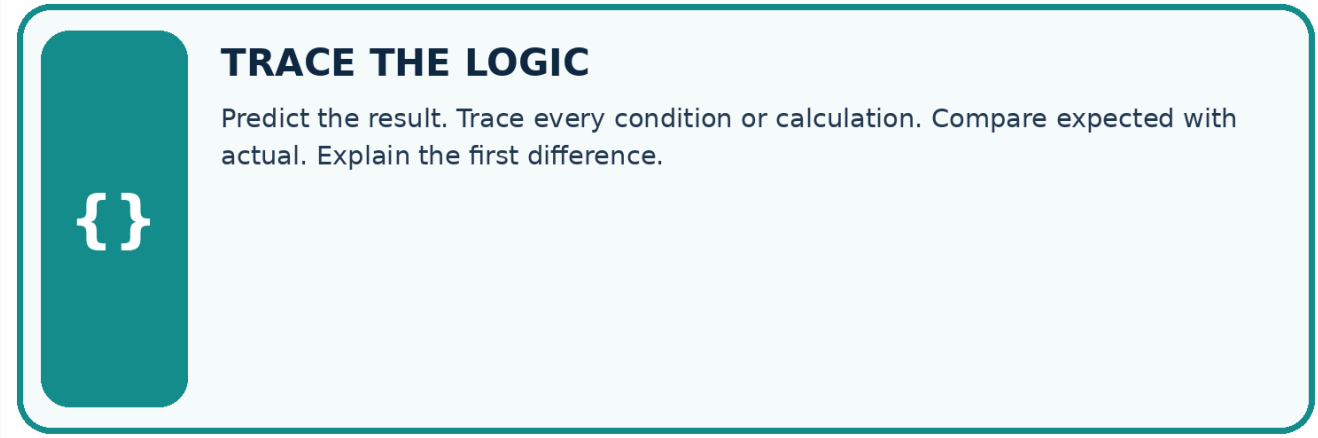
Practise together

> Practise Recommendation Scores with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Choosing the highest **score** and only then ignoring its duration breaks the requirement. Filter for eligibility first. If no item qualifies, report no eligible option instead of relaxing a requirement silently.

Practise with less help: Check a boundary

Use workbook W2 for the connected example “**Weight**ed **score**s and **constraint**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Weight**ed **score**s and **constraint**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why does B not become the winner when its **score** is largest?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

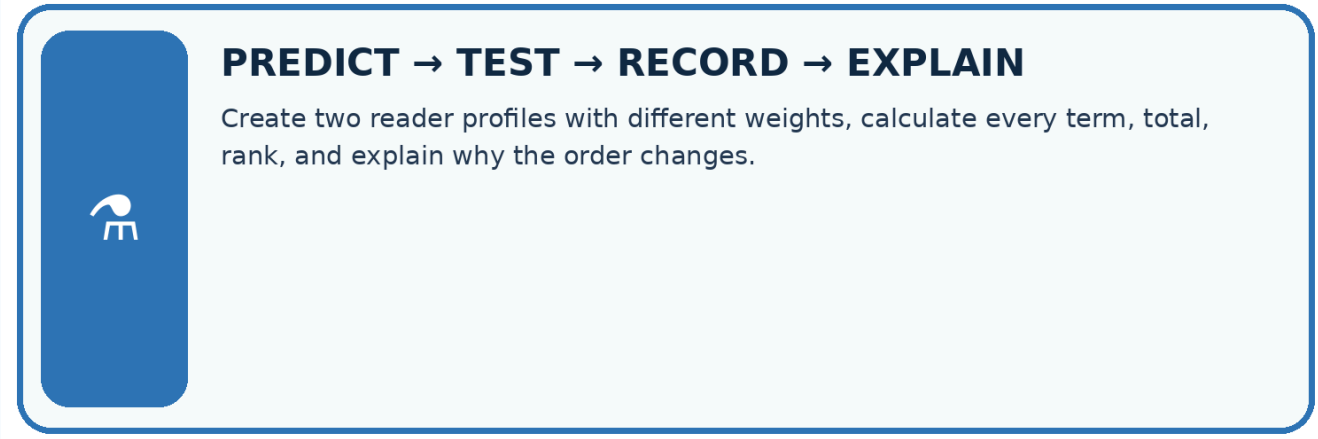
Run the investigation

> Investigate Recommendation Scores. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Create two reader profiles with different **weight**s, calculate every term, total, **rank**, and explain why the order changes.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

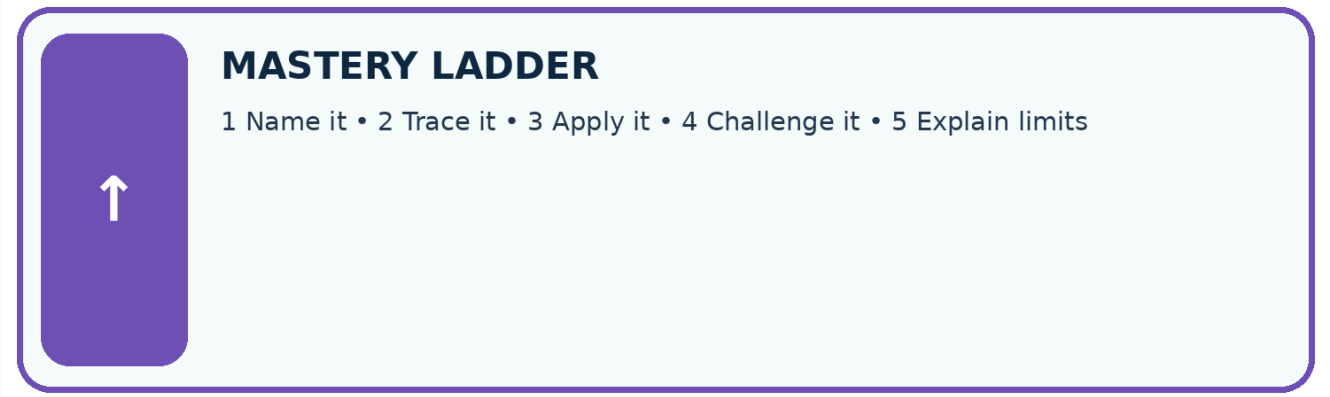
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Recommendation Scores. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

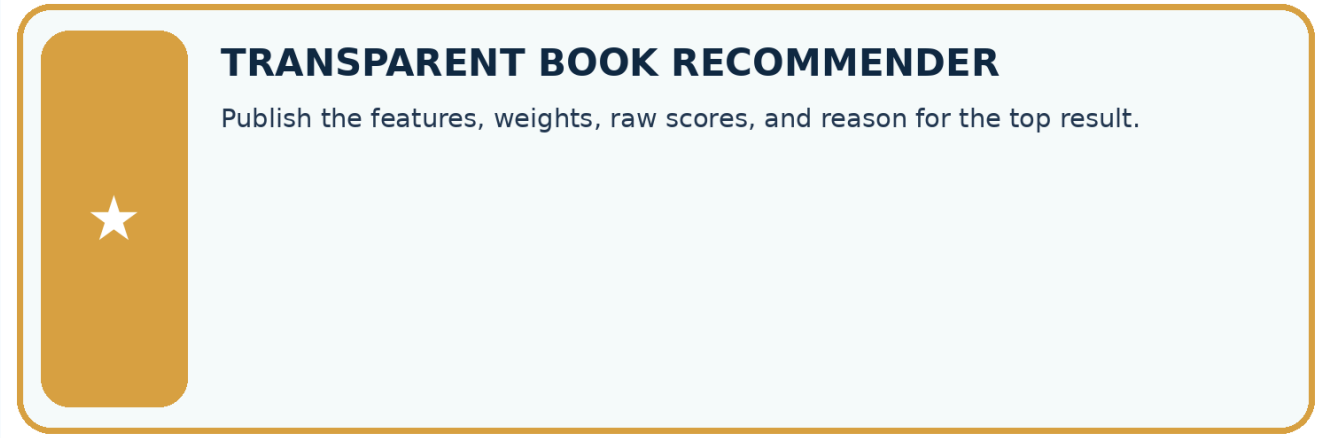
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Publish the features, **weight**s, raw **score**s, and reason for the top result.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or **score** → **rank**ed candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a **rank**ing with a fact about a person.

Evidence for your project

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT PROOF

Create two user profiles and calculate book rankings.

Try a more demanding case

CHALLENGE Create two user profiles and calculate book **rank**ings.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Recommendation Scores case for supported practice. Repeated examples do not become fresh assessments.

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Score=2 times genre match + length match. A matches both, B genre only, C length only.

The question

Score and **rank**.

Worked explanation

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Weighted preferences determine the score

Evidence or stage	Value or result
A	3
B	2
C	1

Score=2 times genre match + length match. A matches both, B genre only, C length only.

Score and **rank**.

A=3, B=2, C=1. **Weight**s encode a chosen preference, not universal reader truth.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Choose independently

Limit is 15 minutes. A has fun 3, learning 4, minutes 15; B has 5, 1, 16; C has 2, 5, 12. Calculate **score**s with 2 * fun + learning and choose.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender gives 3 points for topic match and 1 for difficulty match. Book A matches topic only; B matches difficulty only. Compute **score**s and justify the **rank**ing.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Publish the features, **weight**s, raw **score**s, and reason for the top result.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> **Score**=3×genreMatch+2×short+illustrated. Compute A=[1,0,1], B=[0,1,1], C=[1,1,0]. A reader now requires illustrations. Which book wins after applying that **constraint**?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Recommendation Scores, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional book-activity chooser has a 14-minute limit.

By the end, I can	How I will show it
Compute a weighted score	Explain the deciding step, show a trace or test, and state one limit.
Apply hard constraints before ranking	Explain the deciding step, show a trace or test, and state one limit.
Explain how weight choices change outcomes	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Recommendation Scores without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compute a **weight**ed **score**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-activity chooser has a 14-minute limit.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B's **score** 11 cannot override its failed time requirement.

1. Choose

2. Explain exclusion

3. Eligibility first

4. **Score**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Explain exclusion B's **score** 11 cannot override its failed time requirement.

A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change interest **weight** from 2 to 3, keeping time limit 14. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A **score**s 11; C also **score**s 11; tie-by-ID chooses A. B remains too long; changing preferences changes only **rank**ing of A and C.

Repair a missing building block

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and **rank training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

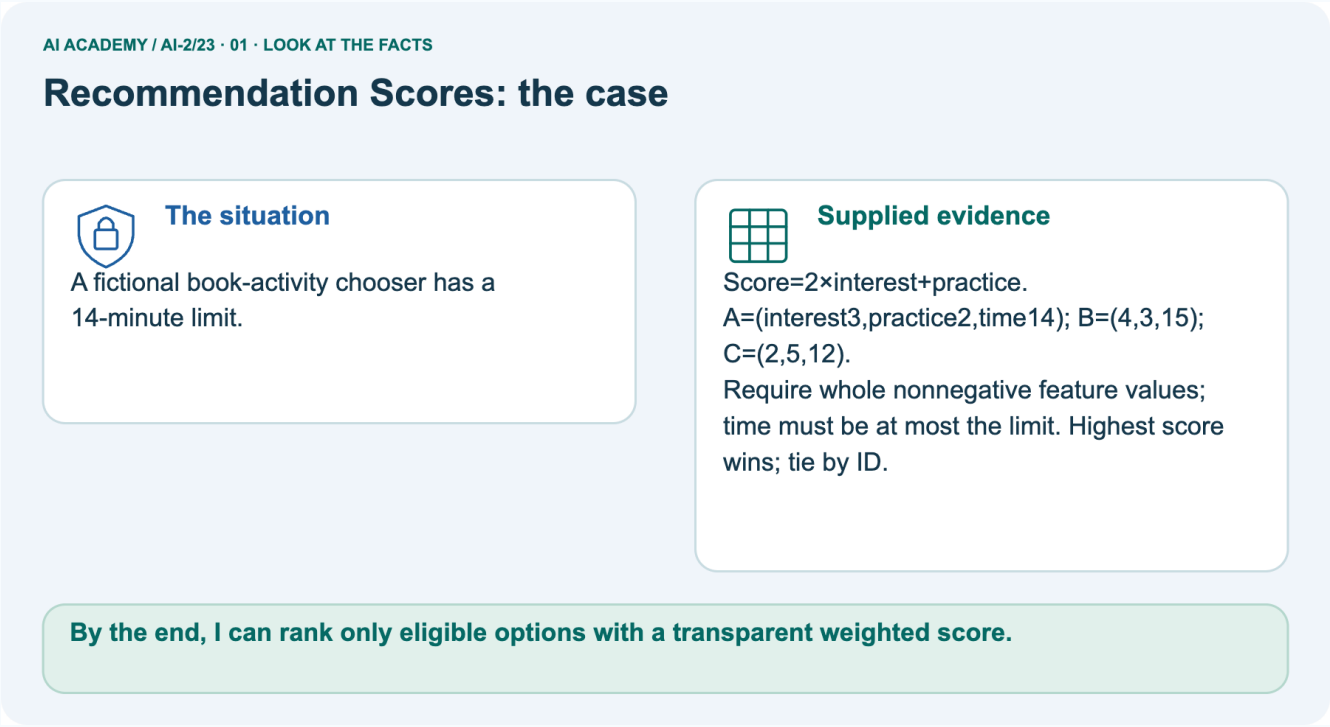
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Recommendation Scores graphic using its labels, deciding evidence and relevant limitation.

01. Recommendation Scores: the case



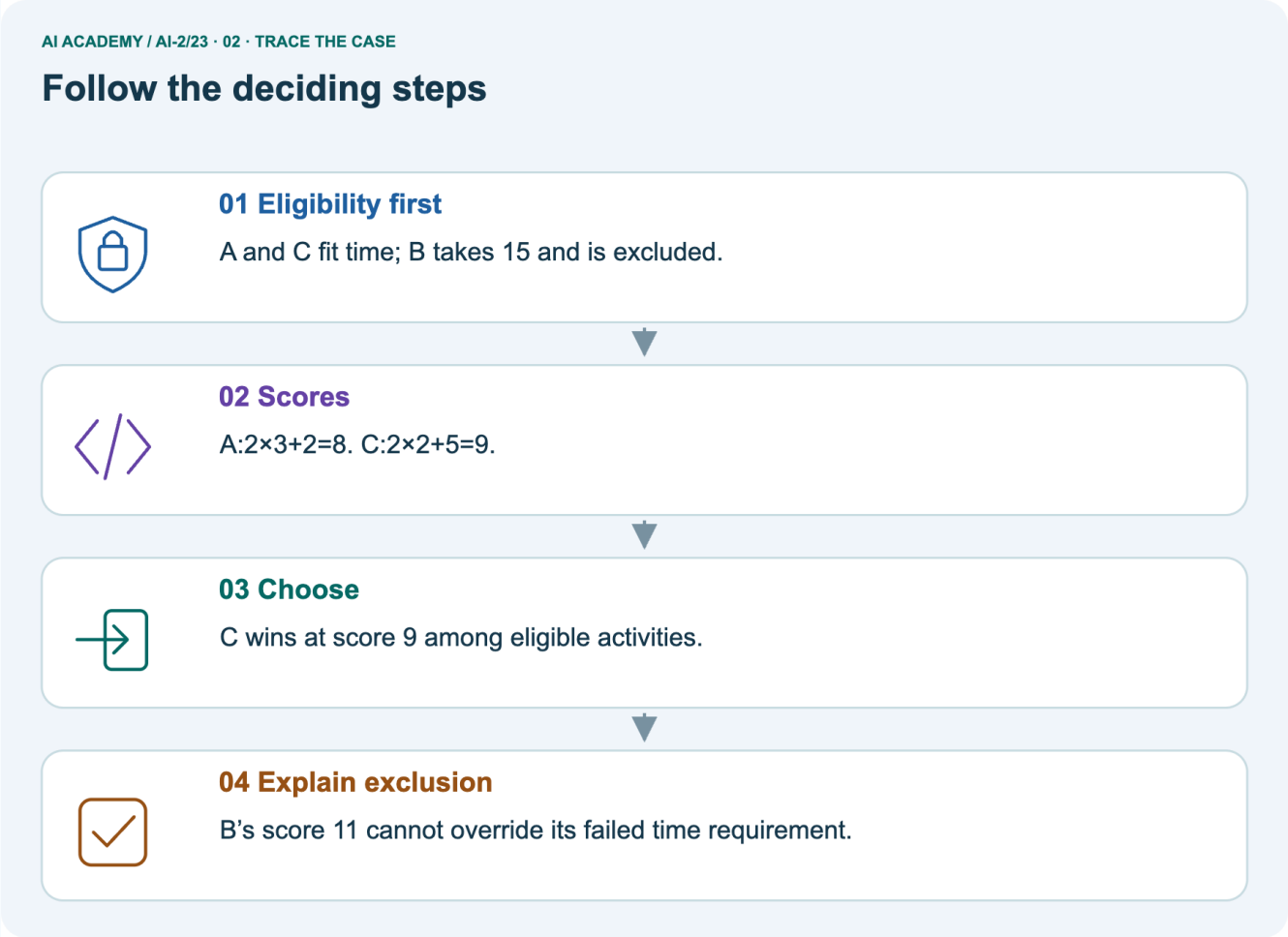
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-activity chooser has a 14-minute limit. Supplied facts: **Score**=2×interest+practice. A=(interest3,practice2,time14); B=(4,3,15); C=(2,5,12).

Require whole nonnegative feature values; time must be at most the limit. Highest **score** wins; tie by ID.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



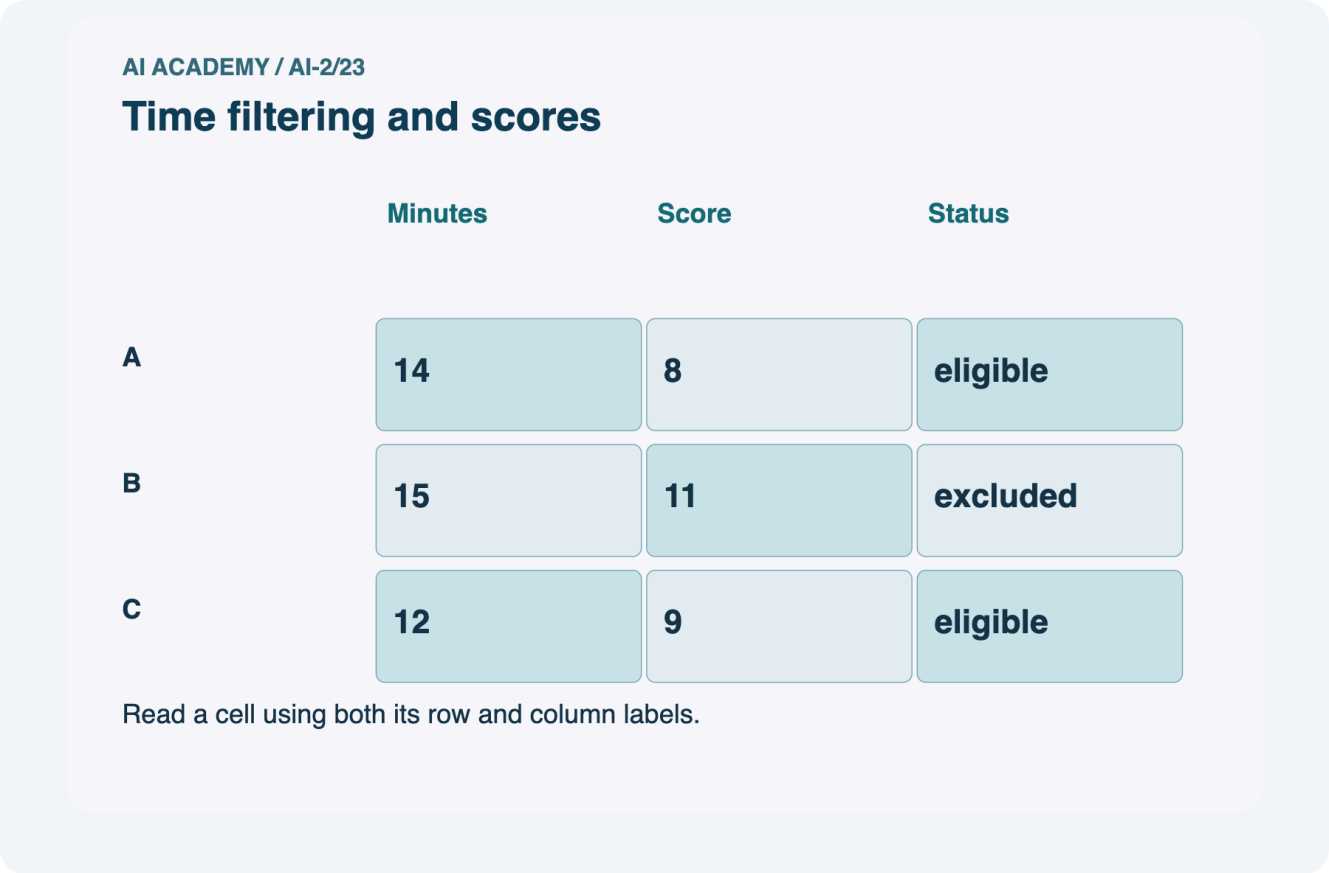
Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility first: A and C fit time; B takes 15 and is excluded. → **Score**s: A:2×3+2=8.

C:2×2+5=9. → Choose: C wins at **score** 9 among eligible activities. → Explain exclusion: B's **score** 11 cannot override its failed time requirement.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Time filtering and scores

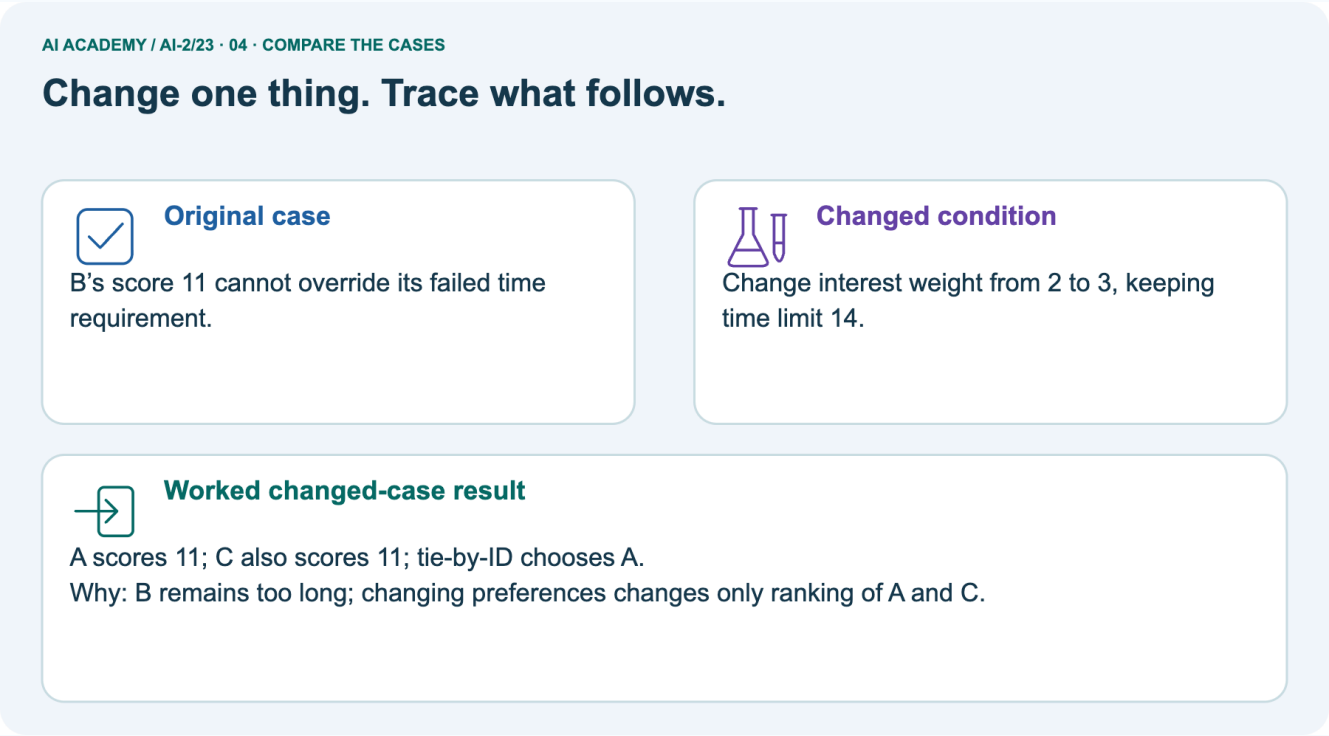


Follow the labelled picture with the matching explanation. Open the vector diagram

Time filtering and **score**s. A: Minutes = 14, **Score** = 8, Status = eligible. B: Minutes = 15, **Score** = 11, Status = excluded. C: Minutes = 12, **Score** = 9, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



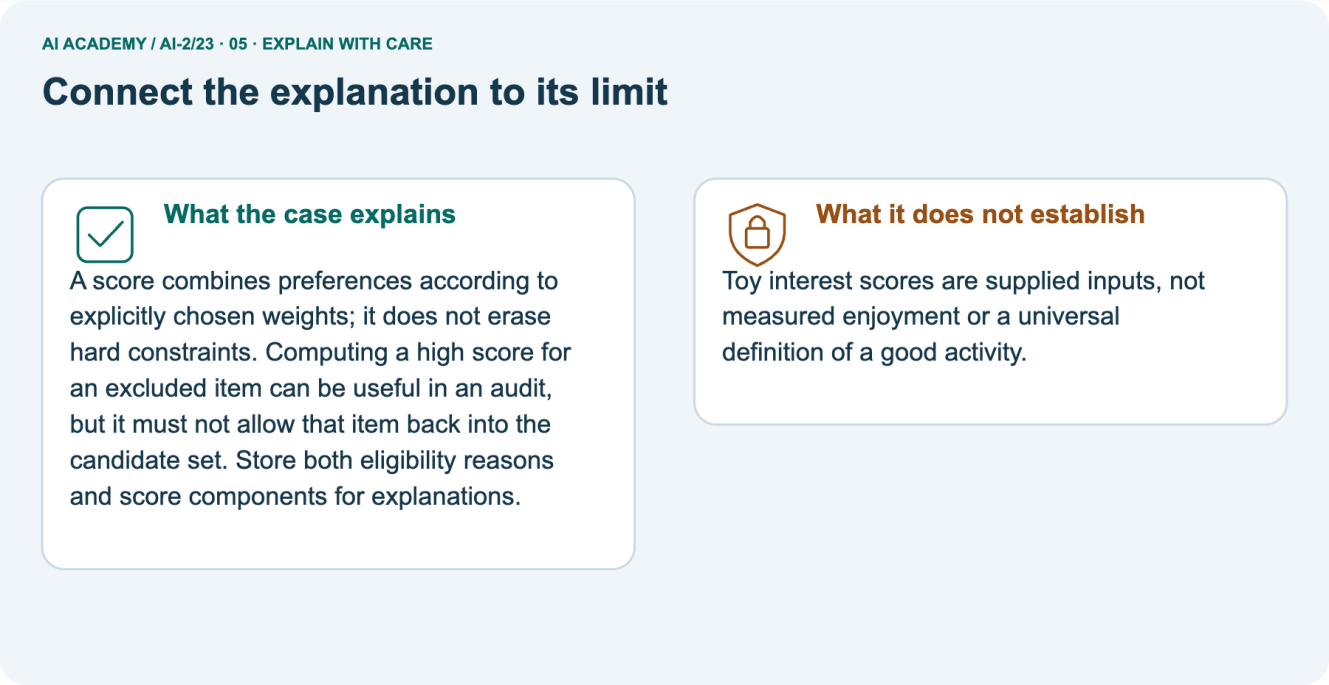
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's **score** 11 cannot override its failed time requirement. Changed condition: Change interest **weight** from 2 to 3, keeping time limit 14. Expected result: A **score**s 11; C also **score**s 11; tie-by-ID chooses A.

Reason: B remains too long; changing preferences changes only **rank**ing of A and C.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

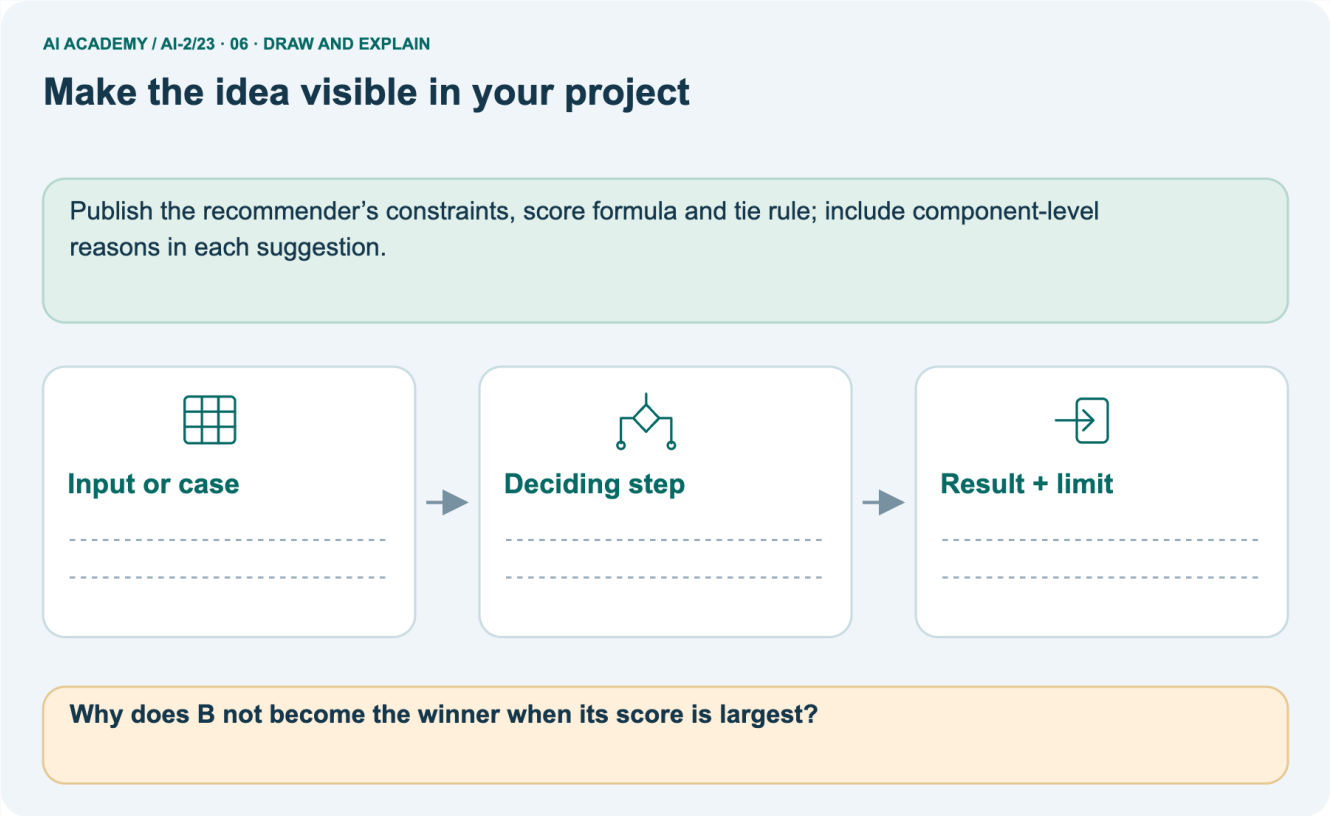
A **score** combines preferences according to explicitly chosen **weight**s; it does not erase hard **constraint**s.

Computing a high **score** for an excluded item can be useful in an audit, but it must not allow that item back into the candidate set. Store both eligibility reasons and **score** components for explanations.

Limit: Toy interest **score**s are supplied **input**s, not measured enjoyment or a universal definition of a good activity.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's **constraint**s, **score** formula and tie rule; include component-level reasons in each suggestion. Draw your **input**, deciding step, result and limit.

Supported question: Why does B not become the winner when its **score** is largest?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.